Experimental Study of PINNs for a Time-Scaled Damped Harmonic Oscillator

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1 Problem Setup

We consider the damped harmonic oscillator:

$$\ddot{x}(t) + 2\xi\dot{x}(t) + x(t) = 0 \tag{1}$$

with:

- $T = 20$
- $\omega_0 = 1$ (time-scaled)
- $\xi \in (0.1, 0.4)$
- $x(0) = 0.7,\ \dot{x}(0) = 1.2$

True solution:

$$x(t) = e^{-\xi t}\left(A\cos(\omega_d t) + B\sin(\omega_d t)\right), \quad \omega_d = \sqrt{1-\xi^2} \tag{2}$$

All references to ω below correspond to ω_{model} (SIREN frequency).

2 Base PINN Configuration

- Architecture: 4 layers $\times$ 32 neurons
- Activations: Tanh, SIREN
- Optimizers:
  - ADAM with varying LR in certain experiments (optimal being 0.001)
  - L-BFGS (1.0) (fixed throughout all the experiments)
- Training: Two phase, first ADAM and then via L-BFGS such that optimal minima may be discovered

Hard constraint formulation via polynomial ansatz:

$$x(t) = x_0 + v_0 t + t^2 N_\theta(t) \tag{3}$$

3 Experiment 1: Damping Ratio Fixed + Tanh vs SIREN

$\xi = 0.3$ is used in this case as well as all the fixed damping experiments uptil Experiment 6.

3.1 Ansatz Forms (exponential and polynomial)

$$x(t) = x_0 + v_0(1 - e^{-t}) + (1 - e^{-t})^2 N_\theta(t) \tag{4}$$
$$x(t) = x_0 + v_0 t + t^2 N_\theta(t) \tag{5}$$

3.2 Results

Model	ω_{model}	Ansatz	Final Loss
SIREN	30	exp	$8.095e^{-1}$
Tanh	-	exp	$8.286e^{-3}$
SIREN	40	exp	$1.384e^{-2}$
SIREN	1	exp	$1.368e^{-5}$
SIREN	30	t^2	$6.66e^{1}$
SIREN	1	t^2	$2.37e^{-1}$
SIREN	0.1	exp	$9.175e^{-3}$

3.3 Observations

- SIREN strongly depends on ω_{model}
- Tanh is more stable but suboptimal for this problem since the solution is periodic
- t^2 ansatz performs poorly
- Best early result: SIREN with $\omega_{\text{model}} = 1$

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4 Experiment 2: Model Frequency Sweep

A search through logspace of model frequency revealed that:

- $\omega_{\text{model}} < 0.4 \rightarrow$ flat loss landscape due to the model capturing less intricacies
- $\omega_{\text{model}} > 1.8 \rightarrow$ rugged landscape as model becomes sensitive to the slightest change in ODE residual loss

Conclusion:

$$0.4 \leq \omega_{\text{model}} \leq 1.8 \text{ is optimal} \tag{6}$$

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5 Experiment 3: Warm Start vs Cold Start

- Warm-start (time marching): slightly worse
- Cold-start at $T = 20$: better convergence

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6 Experiment 4: Loss vs L2 Mismatch

Observed that despite reaching lesser loss levels:

- Loss: 6.36×10^{-7}
- Relative L2: 0.52

Note: Earlier we stumbled upon $\omega_{\text{model}} = 0.611$ as optimal model frequency but on introducing L2 error we find that it is 0.52 or 52 percent which is just the model overfitting the ode loss, henceforth, L2 error is introduced for evaluation.

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7 Experiment 5: Time Horizon Scaling

T	L2	Loss
2.5	$1.04e^{-4}$	$5.1e^{-6}$
5	$1.42e^{-4}$	$1.38e^{-6}$
10	$5.03e^{-3}$	$3.0e^{-5}$
20	$8.49e^{-3}$	$2.16e^{-5}$

** Evaluation performed on a 2k-point linear grid. Training samples were chosen as $160 \times T$ empirically, with scaling applied to avoid sparsity at larger time horizons.*

Observation:

- L2 increases with time
- Loss remains relatively small
- We observe that there is a scale mismatch $\sim 100\times$ between L2 error and ODE loss

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8 Experiment 6: Precise Frequency Sweep (Fixed ξ)

Best observed (ADAM LR 0.001 5k iterations + L-BFGS 3200 iterations):

$$\omega_{\text{model}} \approx 1.137 \tag{7}$$

with:

$$\text{Rel L2} \approx 1.89 \times 10^{-4} \tag{8}$$

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Table 1: Model frequency (ω_{model}) sweep at $T = 20$.

ω_{model}	Loss	Rel L2
0.400	1.48e−06 ± 5.3e−07	4.99e−04 ± 2.6e−04
0.474	1.48e−06 ± 1.4e−06	4.27e−04 ± 3.1e−04
0.547	8.72e−07 ± 3.5e−07	3.30e−04 ± 1.5e−04
0.621	8.09e−07 ± 4.0e−07	3.46e−04 ± 2.3e−04
0.695	1.00e−06 ± 5.8e−07	3.83e−04 ± 2.2e−04
0.768	9.73e−07 ± 4.1e−07	3.22e−04 ± 2.0e−04
0.842	8.08e−07 ± 1.9e−07	2.89e−04 ± 1.7e−04
0.916	9.29e−07 ± 2.3e−07	2.88e−04 ± 1.1e−04
0.989	1.09e−06 ± 2.5e−07	3.35e−04 ± 1.8e−04
1.063	9.02e−07 ± 1.8e−07	2.39e−04 ± 9.1e−05
1.137	1.15e−06 ± 7.9e−07	**1.89e−04 ± 7.6e−05**
1.211	8.69e−07 ± 2.8e−07	2.62e−04 ± 1.3e−04
1.284	7.58e−07 ± 1.4e−07	2.26e−04 ± 7.3e−05
1.358	1.04e−06 ± 3.8e−07	2.63e−04 ± 6.4e−05
1.432	8.70e−07 ± 2.8e−07	2.79e−04 ± 1.0e−04
1.505	9.27e−07 ± 4.2e−07	2.65e−04 ± 1.0e−04
1.579	9.28e−07 ± 3.9e−07	2.81e−04 ± 7.1e−05
1.653	1.13e−06 ± 5.3e−07	2.92e−04 ± 6.3e−05
1.726	9.95e−07 ± 2.9e−07	2.59e−04 ± 3.9e−05
1.800	1.10e−06 ± 2.2e−07	—

9 Experiment 7: Variable ξ

Introducing damping variability (0.1 to 0.4):

- Rel L2 jumps to 6.99×10^{-2}

New frequency behavior:

- Stable range: 0.4 to 4
- Best $\omega_{\text{model}} \approx 3.0787$, as the introduction of variable ξ increases the functional complexity of the solution, requiring higher model frequencies to adequately capture the variations.

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10 Experiment 8: Sample Scaling

- 3k–15k: $\sim 6 \times 10^{-2}$
- 30k: 5.89×10^{-2}
- 60k: 4.9×10^{-2}

Conclusion: diminishing returns, chose 15k.

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11 Experiment 9: Optimization Depth

- L-BFGS improves strongly up to ~ 1600 iterations
- Minimal improvement afterward

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12 Experiment 10: Precision Study

Using float64 (henceforth the norm of future experiments):

$$\text{Rel L2} = 4.32 \times 10^{-3} \tag{9}$$

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13 Experiment 11: Ansatz Formulations

We evaluate multiple hard-constrained ansatz formulations with varying degrees of inductive bias.

13.1 (1) Time-Only Ansatz

$$x(t) = x_0 + v_0(1 - e^{-}t) + (1 - e^{-}t)^2 N_\theta(t, \xi) \tag{10}$$

Exponential decay time ansatz without incorporating damping.

13.2 (2) Exponential Gated Ansatz

$$x(t) = x_0 + v_0(1 - e^{-\xi t}) + (1 - e^{-\xi t})^2 N_\theta(t, \xi) \tag{11}$$

Introduces decay-aware gating aligned with damping dynamics.

13.3 (3) Linear Physics Basis Ansatz

$$x(t) = e^{-\xi t}\left[x_0 + (v_0 + \xi x_0)t\right] + (1 - e^{-\xi t})^2 N_\theta(t, \xi) \tag{12}$$

Uses a first-order physics-informed approximation as the base solution.

13.4 (4) True-Solution-Informed Ansatz

$$\omega_d = \sqrt{1 - \xi^2}, \quad A = x_0, \quad B = \frac{v_0 + \xi x_0}{\omega_d} \tag{13}$$

$$x(t) = e^{-\xi t}\left[A\cos(\omega_d t) + B\sin(\omega_d t)\right] + (1 - e^{-\xi t})^2 N_\theta(t, \xi) \tag{14}$$

Embeds the exact analytical solution, leaving only residual corrections to the network.

13.5 Empirical Comparison

The linear physics basis ansatz performed slightly better than the time-only ansatz, achieving a relative L2 error of 1.9×10^{-3} compared to 2.0×10^{-3}.

Other ansatz formulations did not perform competitively. The true-solution-informed ansatz achieved the lowest error of 1.2×10^{-4}, effectively representing the capacity ceiling of the model, effectively completing relaxing the hypothesis space for the model. We aim in the upcoming experiments to hit 10^{-4} magnitude of L2 error.

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14 Experiment 12: Tanh Architecture with Fourier Features

A Tanh-based architecture was evaluated using the best-performing ansatz, with additional experiments on network depth and width. The optimal configuration was found to be a 4-layer network with 64 neurons per layer.

Fourier feature embeddings were incorporated of the form:

$$\sin(Ax), \quad \cos(Bx) \tag{15}$$

where A and B were sampled from distributions centered above the problem frequency, with standard deviations of 2.0, 1.0, and 0.5.

The feature set dimensionality was also varied across 32, 64, and 128.

Despite extensive tuning across:

- learning rates,
- network depth and width,
- Fourier feature scales and dimensionality,

the model was unable to achieve a relative L2 error below:

$$\text{Rel L2} = 2.9 \times 10^{-2} \tag{16}$$

in any configuration.

Observation: Fourier feature augmentation did not sufficiently compensate for the limitations of the Tanh activation in capturing the oscillatory dynamics of the solution.

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15 Experiment 13: Temporal Weighting

Tested the following to give more weightage to later time steps where the L2 error is higher:

- $e^{\lambda t} \cdot \text{residual}$
- $t^2 \cdot \text{residual}$

Result: worse performance.

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16 Experiment 14: Residual Adaptive Refinement (RAR)

RAR was introduced to improve upon the non-RAR baseline, which achieved:

$$\text{Avg Rel L2} = 4.0697 \times 10^{-3} \pm 2.4 \times 10^{-4} \tag{17}$$

16.1 RAR Setup

RAR was implemented via randomized reseeding, where a subset of training samples was replaced based on high-error regions identified from a pool of 50k evaluated points.

Replacement was applied only during ADAM optimization. L-BFGS was performed on a fixed dataset, ensuring that the loss landscape was not altered during second-stage optimization.

16.2 Results

- Reseeding 3k out of 20k samples (every 400 ADAM iterations over 3k ADAM total, followed by 3200 L-BFGS):
$$3.8661 \times 10^{-3} \pm 1.3 \times 10^{-3} \tag{18}$$
- Replacing 1k samples:
$$2.8041 \times 10^{-3} \pm 3.1 \times 10^{-4} \tag{19}$$
- Replacing 500 samples:
$$2.6131 \times 10^{-3} \pm 4.9 \times 10^{-4} \tag{20}$$
- Replacing 1k samples with increased ADAM iterations (8k):
$$1.3716 \times 10^{-3} \pm 3.5 \times 10^{-4} \tag{21}$$
- Replacing 1k samples more frequently (every 200 iterations, total 3k ADAM):
$$3.4406 \times 10^{-3} \pm 4.4 \times 10^{-4} \tag{22}$$
- Replacing 1k samples less frequently (every 800 iterations, total 8k ADAM):
$$1.5024 \times 10^{-3} \pm 1.8 \times 10^{-4} \tag{23}$$

16.3 Additional Observation

Earlier, global sampling based on high L2 regions was considered. However, isolating those regions and applying ODE residual-based refinement locally proved more effective than optimizing global loss over selected points.

16.4 Key Findings

- Smaller replacement batches (500–1k samples) outperform larger reseeding.
- Less frequent replacement yields better stability and convergence.
- Increasing ADAM iterations (up to 8k) significantly improves performance.

16.5 Projected Behavior

These results suggest that using smaller replacement sets (lesser or equal to 500 samples) with lower replacement frequency and extended ADAM training can push the relative L2 error toward:

$$\text{Rel L2} \to 10^{-4} \tag{24}$$

This aligns with the previously observed lower bound of $\sim 1.2 \times 10^{-4}$ achieved using the true-solution ansatz, indicating that the RAR-enhanced model is approaching the intrinsic capacity limit of the network.

16.6 Additional Observation on Optimization Budget

It is important to note that the RAR experiments were conducted with relatively limited optimization budgets, typically involving 5k to 8k ADAM iterations followed by comparatively fewer L-BFGS iterations.

In contrast, Experiment 11 (ansatz evaluation) utilized significantly larger optimization budgets, with up to 32k L-BFGS iterations, allowing the model to converge much closer to its intrinsic error floor.

Given this discrepancy, it is reasonable to infer that applying RAR under similarly extended optimization (particularly with high L-BFGS iteration counts) would further reduce the error. Specifically, the observed trend suggests that the RAR-enhanced models would approach the previously observed ceiling:

$$\text{Rel L2} \sim 10^{-4} \tag{25}$$

This indicates that the current RAR results are likely optimization-limited rather than capacity-limited.

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17 Conclusion and Future Work

This study demonstrates that PINN performance on the damped harmonic oscillator is primarily governed by the choice of ansatz, model frequency, and adaptive sampling strategy. While standard training approaches plateau around $\mathcal{O}(10^{-3})$, the introduction of RAR and physically-informed ansatz structures enables convergence toward the intrinsic error floor of the model, observed near $\sim 10^{-4}$.

The experiments further indicate that current limitations are largely optimization-driven rather than representational, particularly in the RAR setting where extended optimization budgets were not fully explored.

17.1 Future Work

Further directions considered include:

- Global adaptive sampling strategies were initially explored, but following the effectiveness of RAR, these appeared redundant and were not pursued further.
- Investigation of operator-learning approaches such as DeepONet (branch–trunk architectures) and PINOs (based on Fourier Neural Operators). However, for the current problem setting, these approaches may be unnecessarily complex.

- Exploring alternative evaluation metrics, particularly phase-based error measures, as L2 error alone does not fully capture discrepancies in oscillatory dynamics.
- Stabilizing the relative L2 metric by introducing an ϵ term in the denominator:

$$\frac{|y - \hat{y}|}{|y| + \epsilon} \tag{26}$$

- Revisiting warm-start strategies, which did not perform well in the current setup, but may benefit from improved formulations.
- Further experimentation with higher ADAM learning rates and architectural variations, with the goal of reducing the residual error under the true-solution ansatz into the 10^{-5} to 10^{-6} range.
- Additionally, experimental evidence suggests that using smaller replacement sets ($\leq$ 500 samples), lower replacement frequency, and extended ADAM training (up to 8k iterations) can push the relative L2 error toward the 10^{-4} regime, approaching the previously observed ceiling of $\sim 1.2 \times 10^{-4}$ under maximally optimized settings.